STATISTICAL INFERENCE

Statistical inference is an art of drawing conclusions about the population on the basis of information contained in the sample.

Statistical inference is traditionally divided into main branches

(i) Estimation. **(ii)** Testing the hypothesis.

ESTIMATION:

Estimation is procedure by which we obtain an estimate of the true but unknown value of a population parameter by using sample observations.

There are two types of Estimation.

(i) Point Estimation. **(ii)** Interval Estimation.

(i) POINT ESTIMATION:

Point estimation is a procedure by which we estimate a single value calculated from the sample that is likely to be close to the unknown parameter.

(ii) INTERVAL ESTIMATION:

Interval estimation is a procedure by which we estimate a range of values within which the true value of population parameter is believed to lie with some degree of confidence by using the sample observation.

ESTIMATE AND ESTIMATOR:

An Estimator stands for the rule or method that is used to estimate a parameter whereas an estimate stands for the numerical value obtained by using the rule or method.

e.g If $x_1, x_2, ..., x_n$ is a random sample of size n from a population with mean μ then $\overline{X} = \frac{\sum x}{n}$ is an Estimator of μ and $\overline{x}$ the numerical value of $\overline{X}$ is an estimate of μ.

There are two types of estimates. **(i)** Point estimates. **(ii)** Interval Estimates.

ESTIMATION BY CONFIDENCE INTERVAL:

A confidence interval estimate of the unknown parameter θ is an interval computed from a random sample of size n with a statement of how confident we are that the interval contains the unknown parameter θ.

A confidence interval estimate is in the form (L < θ < U) where L and U depend upon the value of the statistic $\hat{\theta}$ of a random sample selected from the population and the sampling distribution of the statistic.

$$P(L < \theta < U) = 1 - \alpha \qquad \text{for } 0 < \alpha < 1$$

Then the probability of the interval (L, U) containing the population parameter θ is $1-\alpha$. The probability 1 – α is called the confidence coefficient or confidence level. The endpoints that bound the confidence interval are called the lower and upper confidence limits of θ. The difference U – L is called the Precision of the estimate. It is noted that the precision may be increased either by increasing the sample size or by decreasing the confidence level.

UNBIASEDNESS:

An estimator is said to be unbiased if its expected value equal to the true value of the population parameter being estimated. ***e.g.***, if $E\left(\hat{\theta}\right) = \theta$, then $\hat{\theta}$ is an unbiased estimator of the parameter θ otherwise if $E\left(\hat{\theta}\right) \neq \theta$, it is said to be biased estimator.

TESTING OF HYPOTHESIS:

Testing of hypothesis is a procedure which enables to decide on the basis of information obtained by sampling whether to accept or reject any specified statement or hypothesis regarding the value of the parameter in a statistical problem.

STATISTICAL HYPOTHESIS:

A statistical hypothesis is a statement about a characteristic of one or more populations. This statement may or may not be true. Its validity is checked on the basis of information obtained by sampling from the population.

NULL AND ALTERNATIVE HYPOTHESIS:

A hypothesis which is to be tested for possible rejection under the assumption that it is true is called as Null Hypothesis and it is denoted by H_o.

The hypothesis which we accept when the null hypothesis is rejected called an Alternative Hypothesis and it is denoted by H_1.

e.g. Suppose we want to test the hypothesis that the average age of First year college student is 17 Years. Then we formulate the Null and Alternative hypothesis as

H_O: $\mu = 17$ years or $\mu \geq 17$ years or $\mu \leq 17$ years

H_1: $\mu \neq 17$ years or $\mu < 17$ years or $\mu > 17$ years

SIMPLE AND COMPOSITE HYPOTHESIS:

If a hypothesis completely specifies the distribution *i.e.*, it completely specifies the form of the distribution as well as the parameters, it is called a **Simple hypothesis.**

If the hypothesis does not completely specify the distribution, it is called a **Composite hypothesis.**

e.g., Suppose that the age distribution of the first-year college students is normally distributed with mean μ and variance 25. Then H_O: $\mu = 16$ is a simple hypothesis because as the mean and variance together specify a normal distribution completely. On the other hand, if H: $\mu > 16$ (and $\sigma^2 = 25$) or H: $\mu = 16$ (and $\sigma^2 < 25$) it is a composite hypothesis since it does not completely specify the distribution.

TEST STATISTIC:

A Sample statistic which provides a basis for testing a null hypothesis is called a **test statistic.** Every test statistic has its own sampling distribution.

TYPE I AND TYPE II ERRORS:

The rejection of true null hypothesis is called as ***Type I Error***.

The acceptance of false null hypothesis is called as ***Type II Error***.

Hypothesis	Decision	
	Accept H_O	**Reject H_O**
H_O is true	Correct Decision	Type I Error
H_O is false	Type II Error	Correct Decision

The Probability of making Type I error is denoted by α and Type error by β i.e.

$\alpha = P$ (Type I Error)

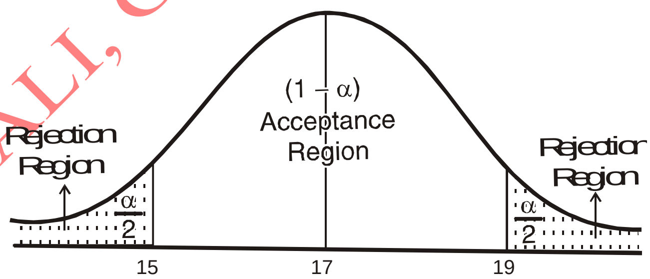
$\beta = P$ (Type II Error)

It is to be noted that when α becomes smaller, β tends to become larger and when α becomes larger, β tends to become smaller. We can decrease both α and β by increasing the sample size.

ACCEPTANCE AND REJECTION REGIONS:

All possible values which a test statistic may assume can be divided into two mutually exclusive groups: one group consisting of values which appear to be consistent with the null hypothesis is called the **acceptance region** and the other having values which are unlikely to occur if H_O is true is called the **rejection or critical region.** The value(s) that separate the critical and acceptance region is called the **critical value(s).**

e.g. Suppose the null hypothesis is that the average age of first year college student is 17 years that is H_o $\mu = 17$ years. It may for example be decided to reject H_o if the sample means $\overline{X}$ is less than 15 years or greater than 19 years. These values (15 and 19) are the critical values of the sample statistic and the area of the sampling distribution of $\overline{X}$ to the right of $\overline{X}$ = 19 and to the left of $\overline{X}$ = 15 is called the critical region or the rejection region. The area of the distribution between $\overline{X}$ = 15 and $\overline{X}$ = 19 is called the acceptance region.

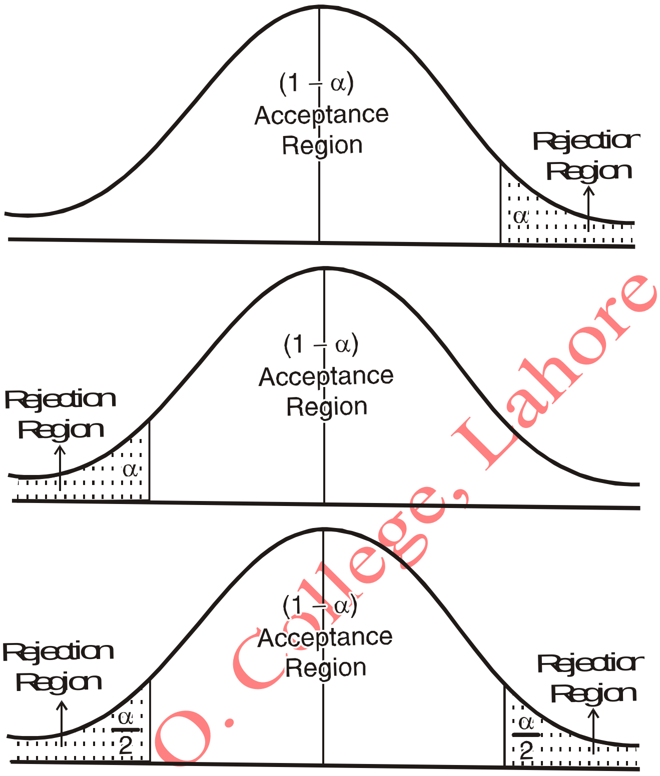


THE LEVEL OF SIGNIFICANCE:

The probability of making a Type I error is called the **level of significance** of the test and is denoted by **α**. In testing a given hypothesis, α is the maximum probability with which we would be willing to risk a Type I error and is generally specified before sample is drawn so that results obtained will not influence our choice.

The most frequently used values of α, the significance level are 0.05(5%) and 0.01(1%). By α = 5% we mean that there are about 5 chances in 100 of incorrectly rejecting a true null hypothesis. To put in another way, we say that we are 95% confident in making the correct decision.

ONE TAILED AND TWO TAILED TESTS:



A test for which the entire rejection region is located in only one of the two tails either in the right tail or in the left tail of the sampling distribution of the test statistic is called a one tailed or one-sided test. A one tailed test is used when the alternative hypothesis H_1 is formulated in the following form

$H_1 : \theta > \theta_0$ or

$H_1 : \theta < \theta_0$

i.e. H_1 is composite hypothesis.

If on other hand the rejection region is divided equally between the two tails of the sampling distribution of the test statistic the test is then called as Two tailed or Two-sided test. In this case the alternative hypothesis is set up as

$H_1 : \theta \neq \theta_0$

i.e. H_1 is two-sided composite hypothesis.

THE USE OF *p-VALUES* IN HYPOTHESIS TESTING:

The *p- value* for a test of hypothesis is defined as the smallest level of significance at which the null hypothesis is rejected or the largest level of significance at which the null hypothesis is accepted.

The *p- value* enables us to test hypothesis without first specifying a value for α. Another benefit of *p- value* is that they provide more information.

For the normal distribution tests, it is relatively easy to compute the P- value. If Z is the computed value of the test statistic, then the P- value is

$$P = \begin{cases} 2\left[1-\phi\left|Z\right|\right] & \text{for a two tailed test} \quad H_0 : \theta = \theta_0 \text{ vs } H_1 : \theta \neq \theta_0 \\ 1-\phi(Z) & \text{for an upper tailed test } H_0 : \theta = \theta_0 \text{ vs } H_1 : \theta > \theta_0 \\ \phi(Z) & \text{for a lower tailed test} \quad H_0 : \theta = \theta_0 \text{ vs } H_1 : \theta < \theta_0 \end{cases}$$

If the P-value is smaller than the significance level, H_0 is rejected and if it is greater than the significance level, H_0 is not rejected.

t-statistic and z-statistic:

if we take a random sample from a normal population with mean μ and variance σ^2, the sample mean $\bar{X}$ is normally distributed with mean μ and variance σ^2/n; that is

$Z = \dfrac{\bar{X}-\mu}{\sigma/\sqrt{n}}$ is a standard normal variable even if n is small provided that σ^2 is known. But in actual practice the population variance σ^2 is usually not known and is estimated from the sample data. So, when sample size n is small (n < 30) is replaced with its unbiased estimate s^2, the statistic

$$t = \frac{\bar{X}-\mu}{s/\sqrt{n}}$$

is called t- statistic and is no longer normally distributed.

Assumption for *t- distribution*:

To use the t-distribution, we make the following assumptions.

(i) The sample of n observations $X_1, X_2, \ldots\ldots, X_n$ is selected randomly.

(ii) The population from which the small samples is drawn is normally.

(iii) In case of two small samples, both the samples are selected randomly, both the populations are normal and both the populations have equal variances.

GENERAL PROCEDURE FOR TESTING HYPOTHESIS:

The procedure for testing a hypothesis about a population parameter involves the following six steps.

1. Formulation of the appropriate null and alternative hypothesis.

2. Decide upon a significance level, α of the test.

3. Choose an appropriate test-statistic.

4. Determination of the rejection or critical region.

5. Calculation of the test statistic.

6. Draw the conclusion.

…………………………………………………..